DiATA RodrisuEZ

10

CHE 313 Transport Phenomena III

Homework #1 (Due:9/11/25, 1:00 pm)

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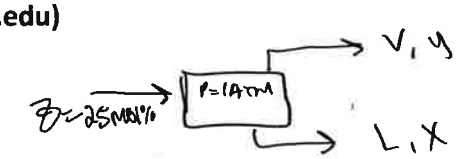


Note: Explain your answers in detail!

1. (3pts) A methanol-water mixture is to be flash distilled at 1 atm. If the feed is 25 mole % methanol, what are the liquid and vapor compositions if:
   a) All of the feed is vaporized.
   b) None of the feed is vaporized.
   c) 1/3 of the feed is vaporized.
   d) 2/3 of the feed is vaporized.

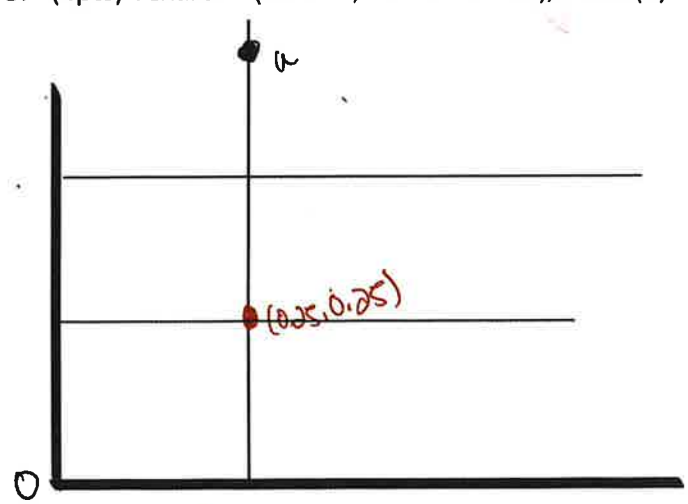
   GRAPH TAble

   Solve all problems (a-d) **graphically** using the information in Table 2-8.

2. (3pts) Flash distillation is used to evaporate half of a 40 mole% methanol-water mixture. The pressure is 1 atm.
   a) What is the composition of the resulting distillate and residue?
   b) By changing the amount of evaporated mixture, what is the maximum possible methanol content of the distillate?
   c) By changing the amount of evaporated mixture, what is the maximum possible water content of the residue?

   Solve all problems (a-c) **graphically** using the information in Table 2-8

3. (4pts) Textbook (Wankat, P. C. 5th edition), 2.D1. (a) – (d)



$\frac{V}{F} = f$

FRACTION Feed vaporized

$y_{n+1} = \frac{F}{V} z = \frac{1}{f} z = \frac{1}{1-q} z$

$y = -\frac{L}{V}x + \frac{F}{V}z$

(2-5) a) All of the feed is vaporized?

$y = 0.25$

$L_{comp} = 0$

$\frac{V}{F} = 1.0 \Rightarrow V = F \Rightarrow \frac{L}{V} = \frac{0}{F} = 0$

0.0411

$(0.045, 0.25)$ ½

$y = -(0)x + 0.25$

b) None of the feed is vaporized

$\frac{V}{F} = 0 \quad V = 0.0 \rightarrow L = F \rightarrow \frac{L}{V} = \frac{F}{0} = \infty$

$x = 0.25$

$y = -\infty x + 0.(0.25)$

$y = -\infty x$

$x = z = 0.25$

$(0.25, 0.62)$ ½

$y = -2x + \frac{1}{\frac{1}{3}}(0.25)$

c) $\frac{V}{F} = \frac{1}{3} \rightarrow V = \frac{1}{3}F$ $\boxed{L = F - V}$

$\frac{L}{V} = \frac{(2/3)F}{(1/3)F} = 2$

$(0.133, 0.484)$

½

$y = -\frac{1}{2}x + \frac{1}{\frac{2}{3}}(0.25)$

d) $\frac{V}{F} = \frac{2}{3} \Rightarrow V\left(\frac{2}{3}\right)F \xrightarrow{L = F - V}$

$L = \left(\frac{1}{3}\right)F$

$\frac{L}{V} = \frac{\left(\frac{1}{3}\right)F}{(2/3)F} = \frac{1}{2}$

$(0.0715, 0.339)$

½

graph ½

a)

V = 0.5 moles

L = 0.5 moles

3

F = Vy + Lx

(1) 0.4 = 0.5y + 0.5x

0.5y → mol frac vapor

0.5x → mol frac liquid

0.8 = y + x => y = 0.8 − x

$x_1 = 0.20$ $y_1 = 0.579$ = 0.779

x = 0.30 y = 0.663 = 0.963

slope $\frac{y_2 - y_1}{x_2 - x_1} = \frac{0.663 - 0.579}{0.30 - 0.20}$ = slope 0.84

$y = y_1 + m \cdot (x - x_1)$

y = 0.579 + 0.84(x − 0.20)

0.8 − x = 0.579 + 0.84(x − 0.20)

0.8 − x = 0.579 + 0.84x − 0.168

0.8 − x = 0.411 + 0.84x

1.84x = 0.389

x = 0.2114

Plus into eq

y = 0.8 − x

y = 0.8 − 0.2114

y = 0.5886

1

x = 0.2114

y = 0.5886

where blue line it crosses

Residue = water

b)

Max possible value of methanol distillate (upper phase)

1

y = 0.72 (blue line) max

from DATA Table 0.72

0.4, 0.724

c) liquid (residue)

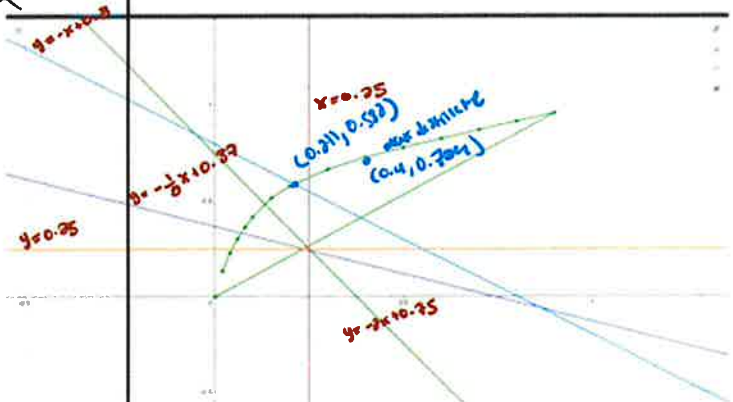
1

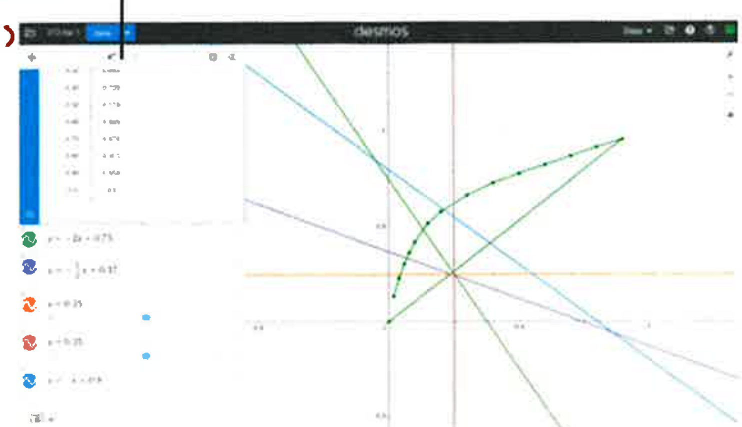
$\frac{V}{F} = 1$ V = 1 L = 0

y = 0.4 → x_{MeOH} = 0.09 min

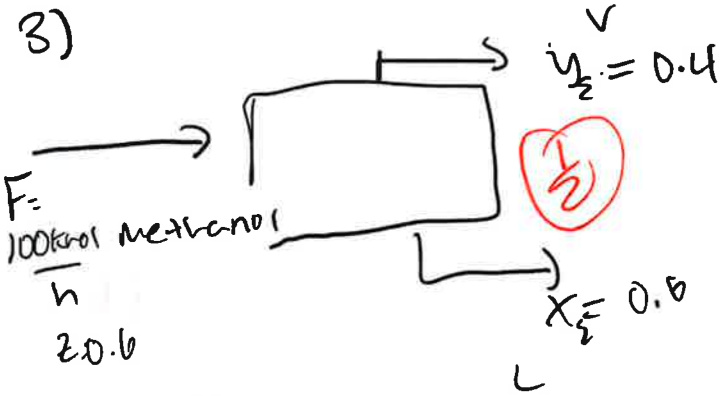
$x_{max, H_2O} = 1 - 0.09$

$x_{max, H_2O} = 0.91$





3)



$$F = L + V$$

$$\frac{V}{F} = 0.4 \qquad V = 0.4\left(100\ \frac{kmol}{h}\right) = 40\ \frac{kmol}{h}$$

$$L = 60\ \frac{kmol}{h}$$

$$Fz = Ly + Vy$$

$$z = 0.6 \qquad y = y_1 + \frac{(x - x_1)(y_0 - y_1)}{(x_0 - x_1)}$$

$$\frac{41.8 - 36.5}{10 - 8} = \frac{40 - 36.5}{x - 8} \qquad 100(0.6) = 60x + 40y \Rightarrow 60 = 60x + 40y$$

$$60 - 60x = 40y$$

$$y = 1.5\% - 1.5x$$

$$y = 72.9 + \frac{(x - 40)(77.9 - 72.9)}{(52 - 40)}$$

$$y = 72.9 + 0.5(x - 40) = 72.9 + 0.5x - 20$$

$$y = 0.5x + 52.9$$

$$0.5x + 52.9 = 150 - 1.5x$$

$$2x = 97.1$$

0.44
$X = 48.55\%$
$x = 0.4855$

$y = 180 - 1.5(48.55) = 77.18$ ①
$y = 0.7718$
0.78
?

$Fz = Lx + Vy$
$(100)(0.6) = 0.6(0.4855) + 40(0.7718)$

b) $\frac{V}{F} = 0.4$ $\quad V = 600 \frac{kmol}{h}$
$F = 1500$

$L = 1500 - 600 = 900 \frac{kmol}{h}$

$y = -\frac{900}{600}x + \frac{1500}{600}z$
$= -1.5x + 2.5(0.6)$
$= -1.5x + 1.5$ ②
$x = 0.4855 \quad y = 0.7718$
0.44

c) $F = V + L \quad Fz = Vy + Lx$
$z = \frac{V}{F}y + \frac{L}{F}x \Rightarrow z = \frac{V}{F}y + (1 - \frac{V}{F})x$
$= \frac{V}{F}y + x - \frac{V}{F}x$
$z = x + \frac{V}{F}(y - x) \rightarrow \frac{z-x}{y-x} = \frac{V}{F}$
$\frac{V}{F} = \frac{0.3 - 0.2}{0.579 - 0.2} = 0.2638$
0.25
$V = 0.2638(1000) = 263.8 \frac{kmol}{h}$ 250 ①
$L = F - V = 1000 - 263.8 = 736.2 \frac{kmol}{h}$ 750
$X = 0.2 \quad y = 0.579$
0.2

d) $x = 0.45 \quad L = 1500 \frac{kmol}{h} \quad V = 0.2(1875)$
$V = 375 \frac{kmol}{hr}$
$F = \frac{L}{0.8} = \frac{1500}{0.8} = 1875 \frac{kmol}{h}$ ①
$y = 72.9 + \frac{(45-40)(77.9-72.9)}{50-40} = 75.9$

$$z = \frac{Lx + Vy}{F} = \frac{(1500)(0.45)+(375)(0.734)}{1875} = 0.5108$$